

Chapter 2

Digital Image Fundamentals

2.6 Mathematical Tools Used in Digital Image Processing

- 2.6.1 Array/Matrix
- 2.6.2 Linear/Nonlinear Operation

$$H[f(x, y)] = g(x, y)$$

$$H[a_i \cdot f_i(x, y) + a_j \cdot f_j(x, y)] = a_i \cdot H[f_i(x, y)] + a_j \cdot H[f_j(x, y)]$$

Exercise:

Give two examples of linear operations & two examples of nonlinear operations. (input is 1-D signal $f(x)$)

- 2.6.3 Arithmetic Operation

$$s(x,y) = f(x,y) + h(x,y)$$

$$d(x,y) = f(x,y) - h(x,y)$$

$$p(x,y) = f(x,y) \times h(x,y)$$

$$v(x,y) = f(x,y) \div h(x,y)$$

Application of Arithmetic Operation

(1) Addition

$$g(x,y) = f(x,y) + n(x,y)$$

$n(x,y)$ is noise image.

Emulate Noise Effect

[Programming: `Rand()`, Saturation Operation]

Application of Arithmetic Operation

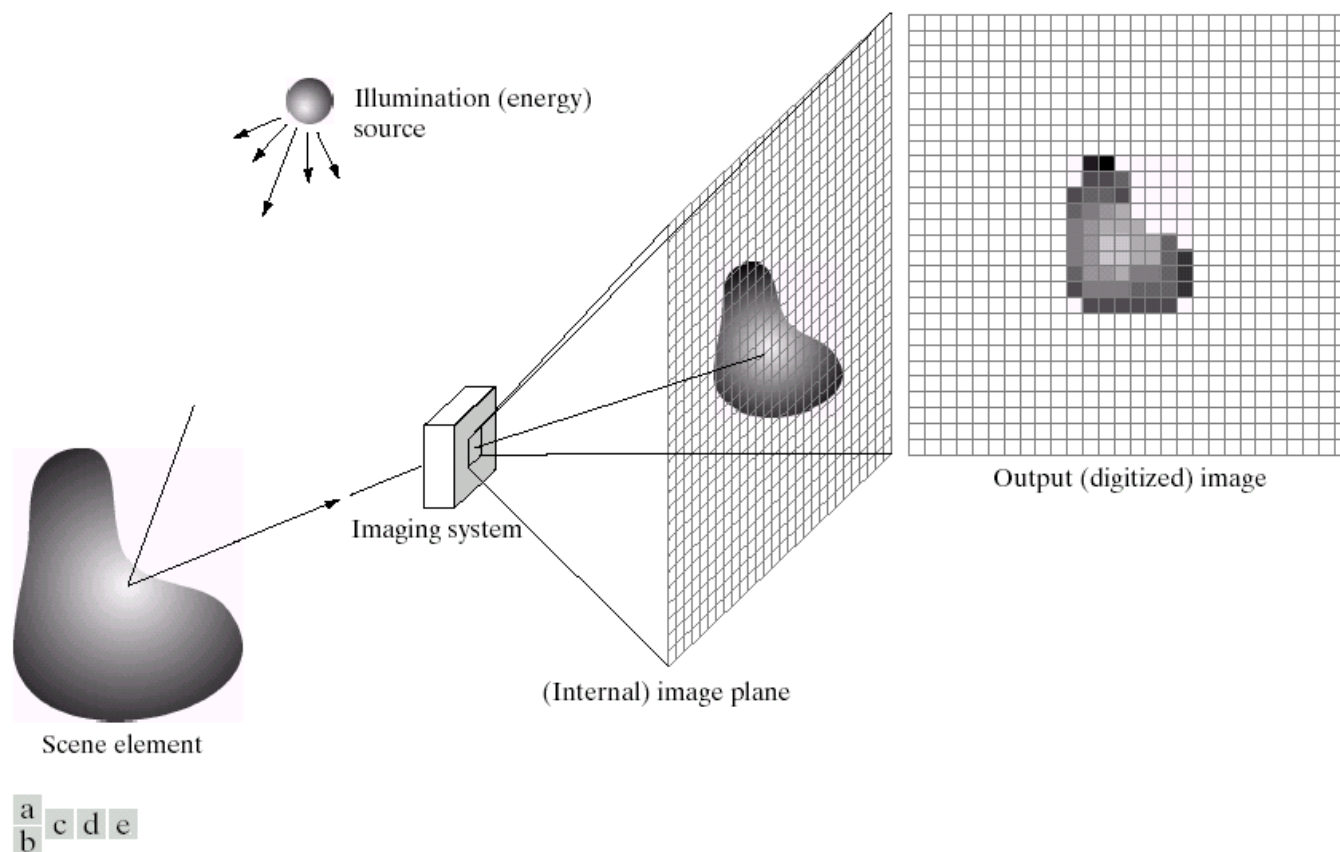
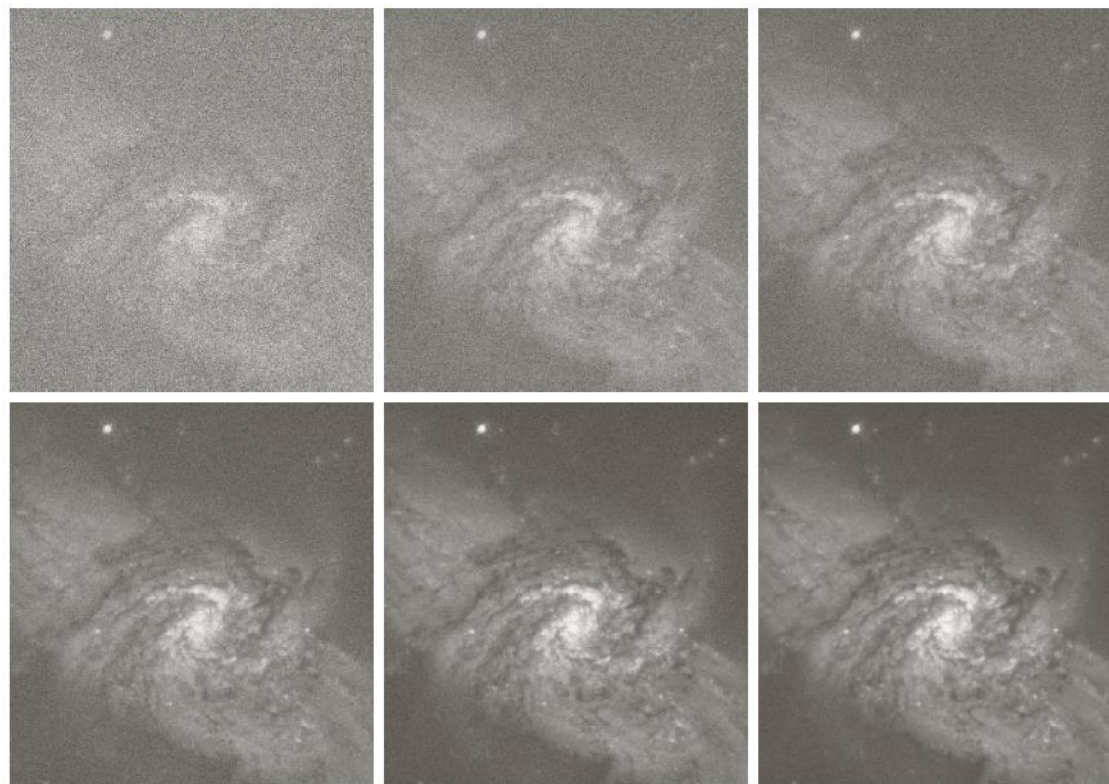


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

- Averaging K different noisy images

$$g_i(x, y) = f(x, y) + n_i(x, y)$$

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$



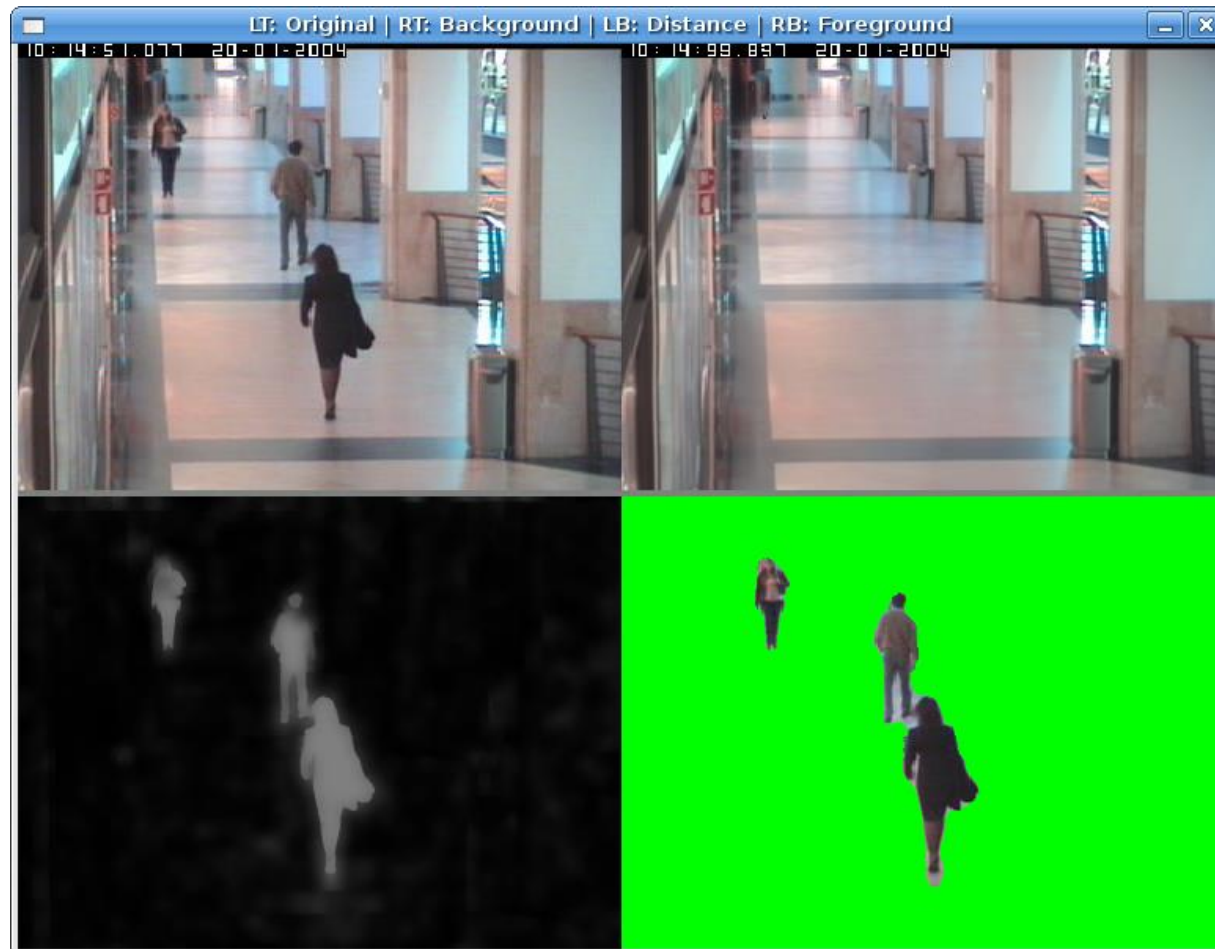
a	b	c
d	e	f

FIGURE 2.26 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)–(f) Results of averaging 5, 10, 20, 50, and 100 noisy images, respectively. (Original image courtesy of NASA.)

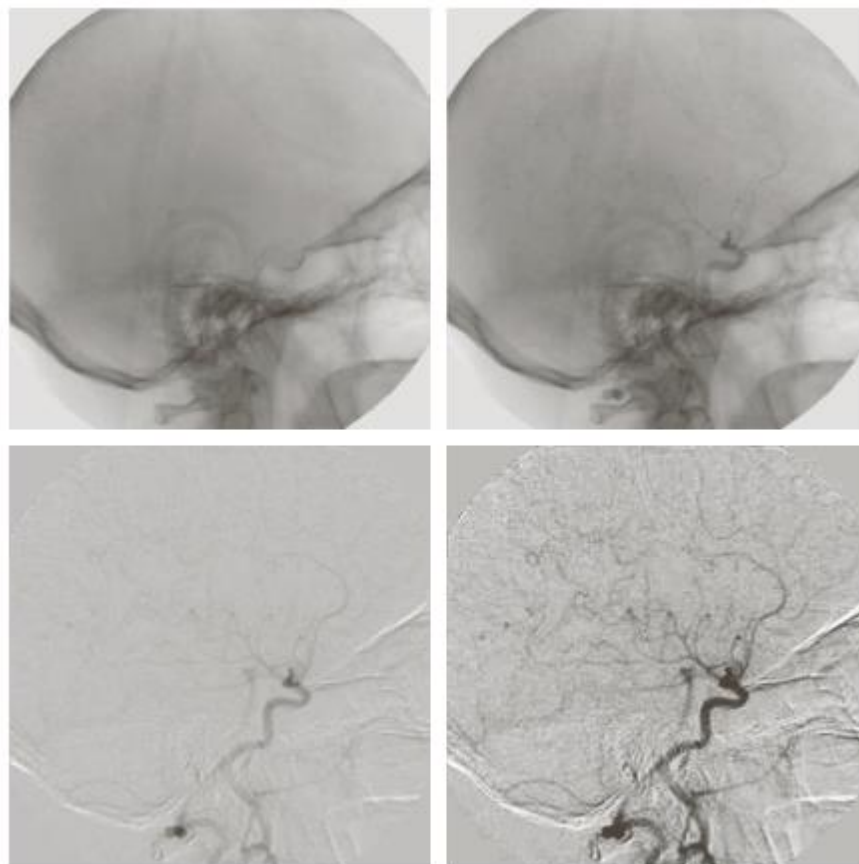


- Subtraction
 - $d(x,y) = f(x,y) - h(x,y)$
 - Enhance the difference between 2 images

Application of image subtraction 1



Application of image subtraction 2



a	b
c	d

FIGURE 2.28

Digital
subtraction
angiography.

(a) Mask image.

(b) A live image.

(c) Difference
between (a) and
(b). (d) Enhanced
difference image.

(Figures (a) and
(b) courtesy of
The Image
Sciences Institute,
University
Medical Center,
Utrecht, The
Netherlands.)

- Multiplication

$$g(x,y) = f(x,y) h(x,y)$$

- $f(x,y) = i(x,y) r(x,y)$,Image Formation Model, $i(x,y)$: illumination components, $0 < i(x,y) < \infty$, $r(x,y)$: reflectance components, $0 < r(x,y) < 1$

Emulate lighting effect.

$$g(x,y) = f(x,y) l(x,y)$$

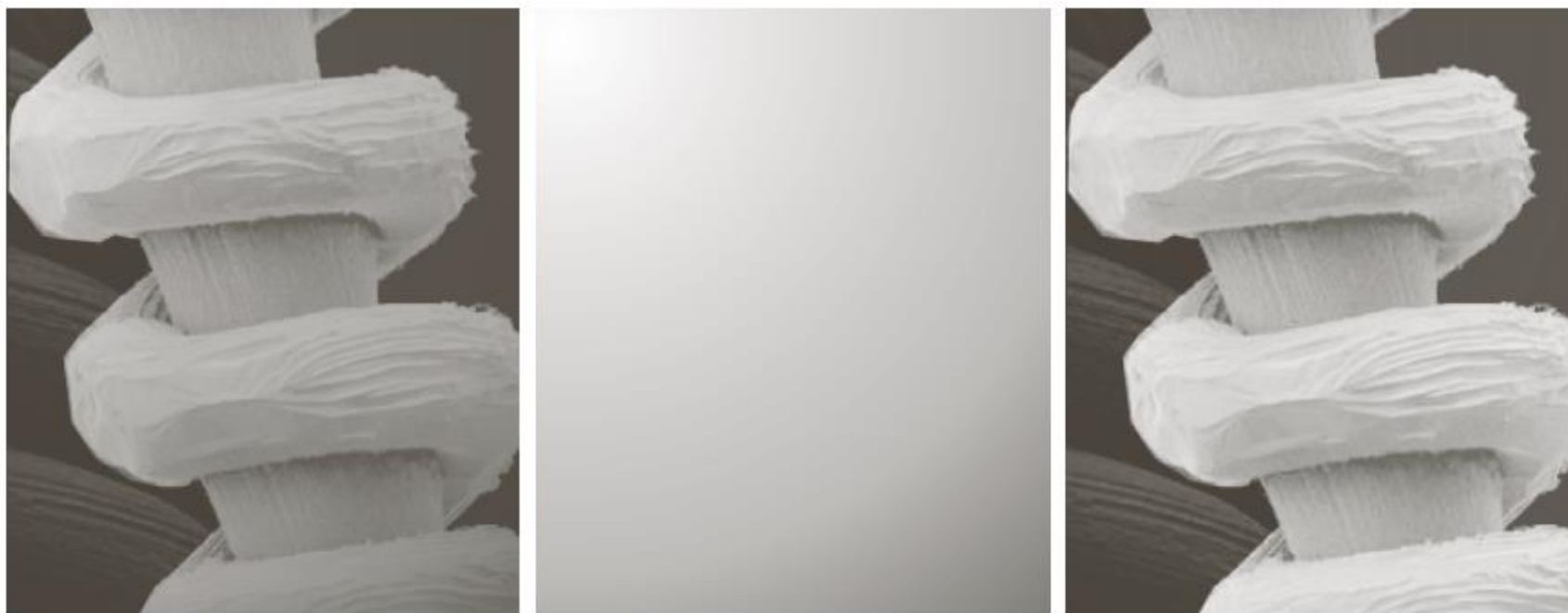
- Division
 - Shading Correction

$$f(x,y) = g(x,y) / I(x,y);$$

or Multiplication

$$f(x,y) = g(x,y) * (1/I(x,y))$$

Application of image multiplication or division



a b c

FIGURE 2.29 Shading correction. (a) Shaded SEM image of a tungsten filament and support, magnified approximately 130 times. (b) The shading pattern. (c) Product of (a) by the reciprocal of (b). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

Application of Image Multiplication ROI (region of interest) operation



a b c

FIGURE 2.30 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

Other Application of Arithmetic Operation

- How to display a image $f(x,y)$ which value range is out of $[0..255]$

$$f_m = f - \min(f);$$

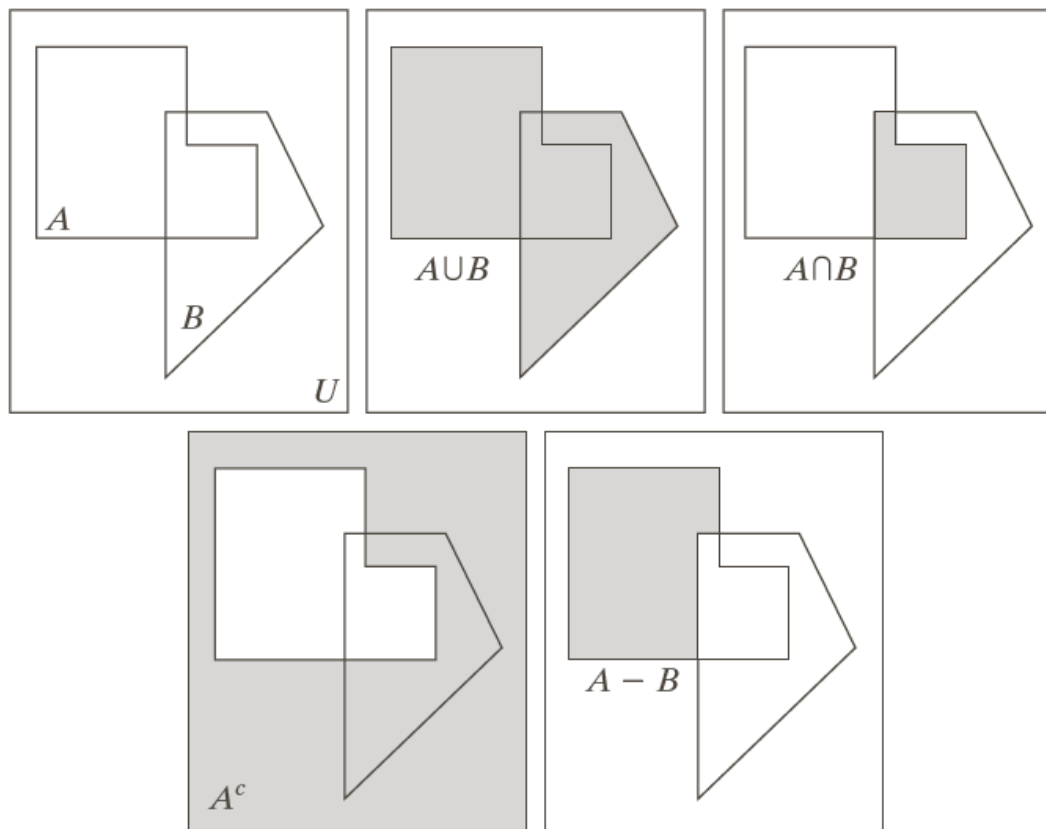
$$f_s = K(f_m / \max(f_m)), \text{ where } K = 255$$

Mix Operation

- $g(x,y) = f1(x,y)*(1-u)+f2(x,y)*u$

2.6.4 Set And Logic Operation

(1) Set Operation



2.6.4 Set And Logic Operation

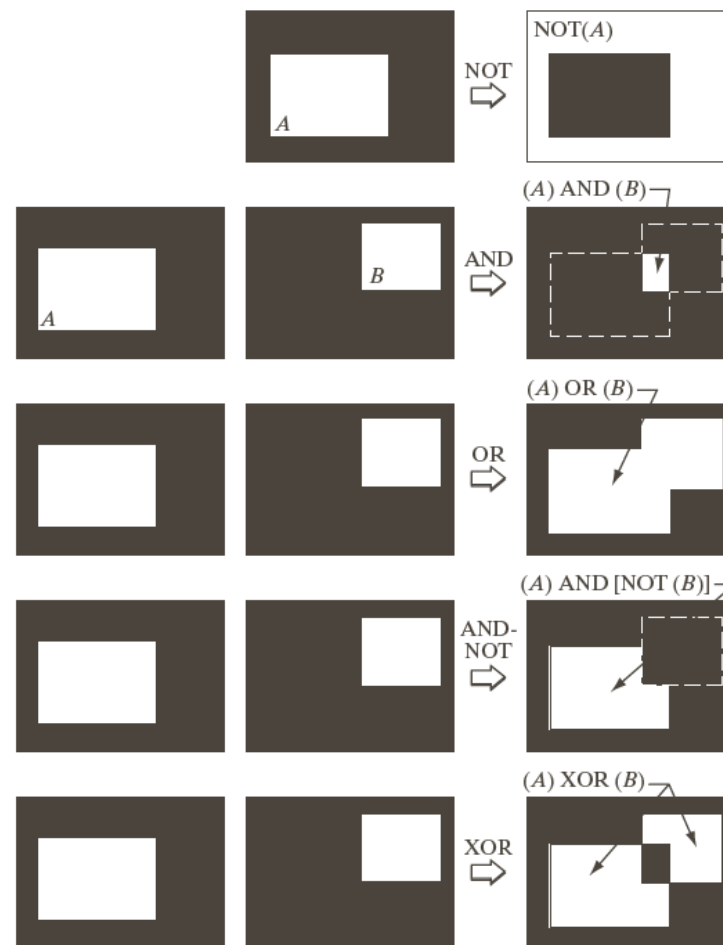
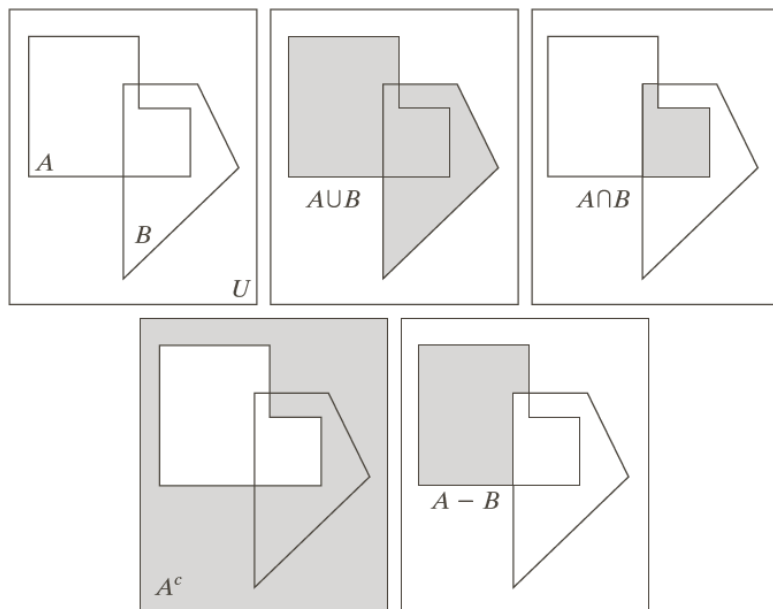
(2) Logic Operation



FIGURE 2.33

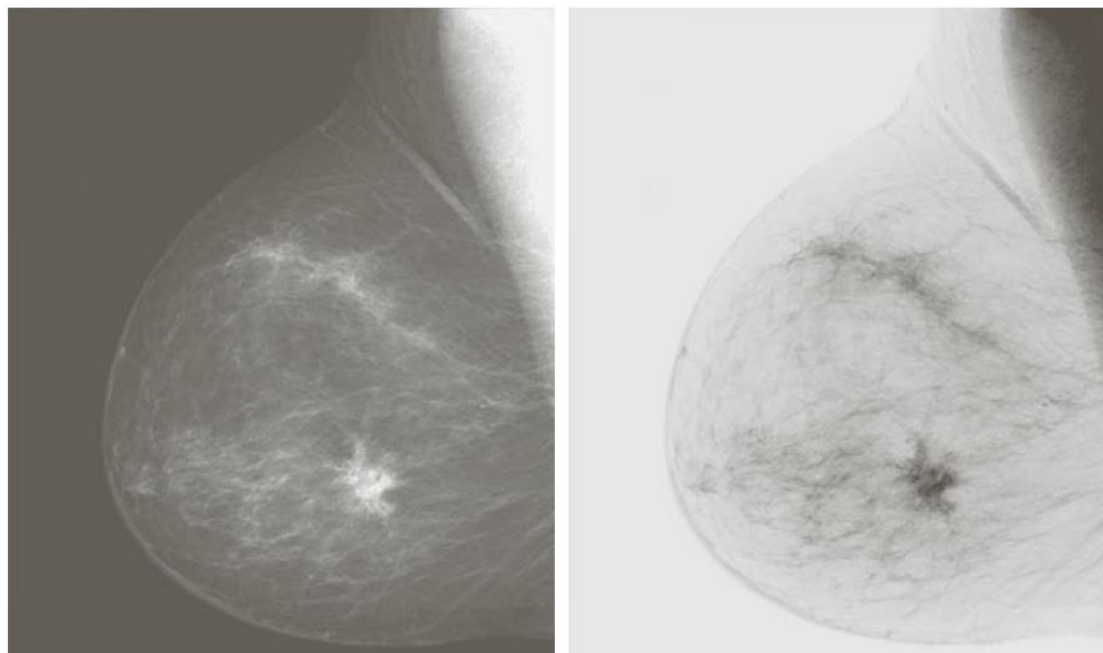
Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.

Set Operation / Logic Operation



2.6.5 Spatial Operation

- (1) Single pixel operation: $g(x,y) = T(f(x,y))$
 $g[x,y] = L-1-f[x,y]$; L is gray levels (256)



2.6.5 Spatial Operation

- (2) Neighborhood operation

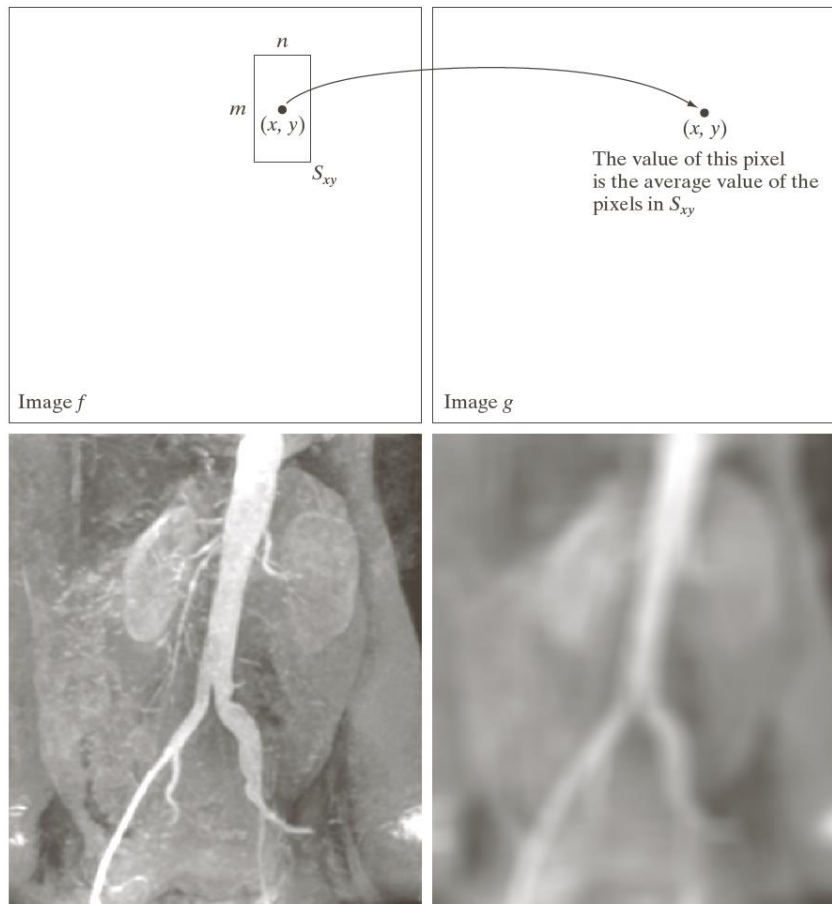


FIGURE 2.35

Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) The aortic angiogram discussed in Section 1.3.2. (d) The result of using Eq. (2.6-21) with $m = n = 41$. The images are of size 790×686 pixels.

Neighborhood operation

- Local averaging using 8-neighbors

$$g[x,y] = (f[x,y] + f[x-1,y] + f[x+1,y] + f[x,y-1] + f[x,y+1] + f[x-1,y-1] + f[x+1,y-1] + f[x-1,y+1] + f[x+1,y+1]) / 9$$

Exercise

- A: $g[x,y] = f[x,y] - f[x-1,y];$
- B: $g[x,y] = f[x,y] + 10;$

2.6.5 Spatial Operation

- (3) Geometric Image Transform

Spatial Transform

2.6.6 Vector and Matrix Operations

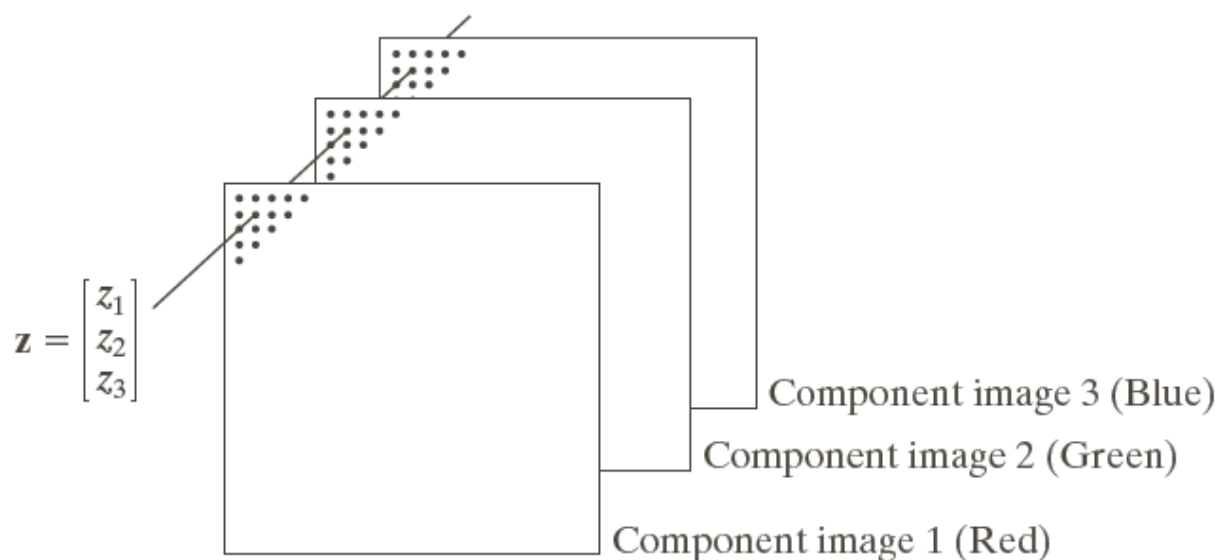


FIGURE 2.38

Formation of a vector from corresponding pixel values in three RGB component images.

2.6.7 Image Transform

- Forward Transform

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v)$$

- Inverse Transform

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) s(x, y, u, v)$$

2.6.7 Image Transform

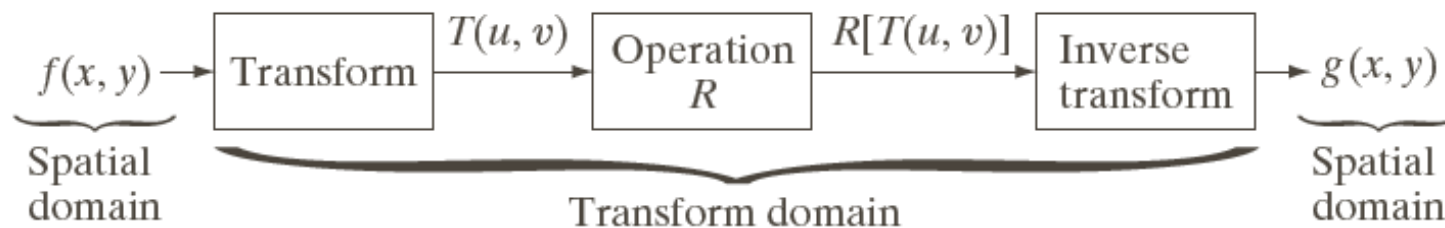
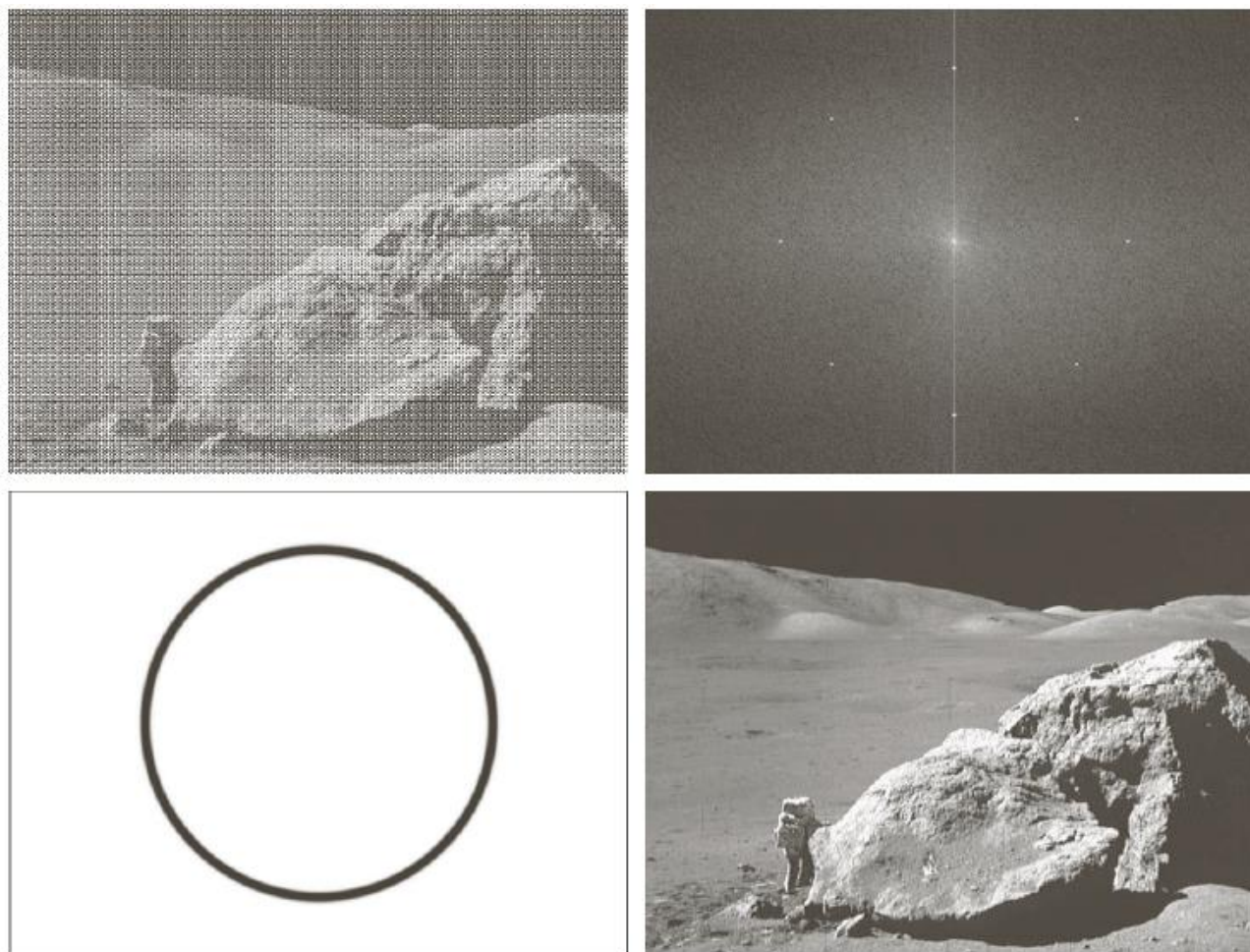


FIGURE 2.39

General approach
for operating in
the linear
transform
domain.

2.6.7 Image Transform



a	b
c	d

FIGURE 2.40

(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy responsible for the interference. (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)